MicroPsi2, Psi Theory, and the State of Machine Consciousness

A Survey of Architectures for Motivated Cognition and Artificial Inner Life

Jung Agent Project

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Abstract

This survey examines the theoretical foundations and software architecture of MicroPsi2, Joscha Bach's implementation of Dietrich Dörner's Psi theory of motivated cognition. We situate MicroPsi2 within the broader landscape of machine consciousness research, surveying Global Workspace Theory, Integrated Information Theory, Attention Schema Theory, predictive processing, and higher-order theories. We then assess the alignment between the Jung Agent—a Jungian archetypal cognitive architecture for situated AI—and the Psi theoretic tradition, identifying both convergences and divergences that inform the project's future direction.

Contents

1 Introduction

The question of whether machines can possess inner life—not merely simulate behavior, but *experience* states—has moved from philosophical speculation to engineering concern. As AI systems grow more autonomous, embodied, and socially situated, the architectures that govern their motivational dynamics increasingly resemble the very cognitive models proposed to explain human consciousness.

Psi theory, developed by Dietrich Dörner beginning in the 1990s and formalized computationally by Joscha Bach as MicroPsi and later MicroPsi2, offers one of the most complete frameworks for building agents whose behavior arises from homeostatic drives, emergent emotions, and modulated cognitive processing (Dörner, 1999; Bach, 2009). Unlike classical cognitive architectures such as Soar (Laird, 2012) or ACT-R (Anderson et al., 2004), which focus on problem-solving and production rules, MicroPsi2 places motivation and emotion at the *center* of the architecture rather than treating them as peripheral phenomena.

This survey has three aims:

1. Present the architecture of MicroPsi2 and its theoretical foundations in Psi theory with sufficient depth for a technical audience.
2. Survey the current state of the art in machine consciousness research, identifying the major theoretical frameworks and their computational implications.
3. Assess where the Jung Agent project—a Jungian archetypal cognitive architecture for macOS—sits within this landscape.

2 Psi Theory: Foundations

2.1 Overview

Psi theory, developed by Dörner at the University of Bamberg, is a systemic psychological theory covering human action regulation, intention selection, and emotion (Dörner & Güss, 2013). It models the human mind as an information-processing agent controlled by a set of basic *physiological*, *social*, and *cognitive* drives. Perceptual and cognitive processing are directed and modulated by these drives, which allow the autonomous establishment and pursuit of goals in an open environment.

Key inspirations for Psi theory include the TOTE (Test-Operate-Test-Exit) model from Miller, Galanter, and Pribram (1960) and biological concepts of homeostasis that underscore the maintenance of internal balance against external perturbations.

2.2 The Drive System

Drives in Psi theory are homeostatic regulators. Each drive has a *setpoint* (desired level), a *current level*, and the *demand* signal is the deviation between them. When demand is high, the drive becomes urgent and dominates behavior selection.

Dörner identifies three categories of drives. Note that certainty operates in both social and cognitive domains—Dörner treats *legitimacy* (predictability in the social order) and *epistemic certainty* (resolved uncertainty about the world) as related but distinct regulatory needs:

Table 1: Drive taxonomy in Psi theory (Dörner & Güss, 2013)

Category	Drive	Description
Existential	Energy	Metabolic resources, sustenance
	Integrity	Physical safety, structural wholeness
	Arousal	Optimal stimulation level (avoiding boredom and overwhelm)
Social	Affiliation	Need for social connection, belonging
	Legitimacy	Need for predictability in the social order
Cognitive	Competence	Need for mastery, control over environment
	Certainty	Need for predictability, resolved uncertainty

The Jung Agent extends this taxonomy with three additional drives: curiosity (explicit exploratory motivation, implicit in Psi's competence/certainty interaction), recognition (explicit social acknowledgment, subsumed within Psi's social drives), and individuation (a Jungian contribution: long-term growth toward psychological integration, absent from Psi theory entirely). These extensions are analyzed in detail in the companion competitive analysis.

Drive dynamics produce two critical signals:

- Pleasure: experienced when a drive's demand *decreases* (the satisfaction signal).
- Pain/displeasure: experienced when a drive's demand *increases* (frustration or deprivation).

These signals do not require a conscious observer—they are functional states that modulate behavior selection, making Psi theory compatible with both functionalist and embodied accounts of affect.

2.3 The Modulator System

Perhaps the most distinctive feature of Psi theory is its *modulator* layer—a set of parameters that shape *how* the agent thinks, not merely *what* it wants. Modulators derive from drive states and in turn influence cognitive processing:

Table 2: Psi modulators and their cognitive effects

Modulator	Derivation	Cognitive Effect
Arousal level	Overall drive demand intensity	High → broad but shallow processing; low → narrow and deep
Resolution level	Inverse of arousal; certainty-related	High → fine-grained discrimination; low → coarse categorization
Selection threshold	Success/failure history	High → only strong signals trigger action; low → impulsive, easily triggered
Securing rate	Competence satisfaction	High → thorough verification of outcomes; low → quick acceptance, moving on
Goal valuation	Anticipated vs. experienced satisfaction	Bias toward novel goals vs. repeating known-satisfying ones

2.4 Emergent Emotions

In Psi theory, emotions are *not* discrete states to be tracked or triggered. They *emerge* from modulator configurations:

Emotion as Modulator Configuration

Anger = high arousal + low resolution + competence frustration
Fear = high arousal + high certainty demand + integrity threat
Joy = low arousal + high resolution + recent drive satisfaction
Sadness = low arousal + low securing rate + high affiliation demand
Curiosity = moderate arousal + high resolution + low certainty

This approach is more powerful than discrete emotion models because it produces a continuous, high-dimensional emotional landscape from a small number of parameters. Emotions are not *recognized*—they are *lived* as the current configuration of the modulator space (Bach, 2012).

2.5 Working Memory and Anticipation

The working memory in Psi theory is conceptualized as a *protocol chain* consisting of:

- The image of the current situation
- The *expectation horizon*—a branching tree of anticipated future developments and active plans
- The remembered past
- The intention memory

The expectation horizon provides forward modeling: the agent simulates potential future scenarios to minimize surprise and optimize drive satisfaction. This anticipatory control loop connects Psi theory to modern predictive processing frameworks (Friston, 2010).

2.6 Action Regulation

Action atoms in Psi theory are triplets: a partial situation description (condition), an operator (action), and an expected outcome. These triplets can be chained into hierarchical plans and stored as *protocol chains* in memory. Decay of protocol chains follows evolutionary principles—chains that lead to drive satisfaction are strengthened, others fade.

3 MicroPsi2: Architecture

3.1 From Theory to Software

MicroPsi2 is Joscha Bach's second-generation implementation of Psi theory as a computational cognitive architecture (Bach, 2012). It translates Dörner's psychological theory into a running software framework for building autonomous agents.

3.2 Node Nets

The central representational substrate in MicroPsi2 is the *node net*: a semantic network with spreading activation that serves as a unified representation for control structures, plans, sensory and motor schemas, Bayesian networks, and neural nets (Bach, 2009).

Node types include:

- Concept nodes: represent categories, objects, situations
- Sensor nodes: interface with the environment (input)

- Motor nodes: interface with actuators (output)
- Activation nodes: carry spreading activation signals
- Register nodes: hold temporary state for computation

Spreading activation propagates through the net according to link weights, with modulator values influencing the activation function globally. High arousal, for example, increases the breadth of activation spread (more associations are activated) while decreasing depth (each individual activation is weaker).

3.3 World Adapters

MicroPsi2 connects to environments through *world adapters*—plugins that map between the agent's sensor/motor nodes and external systems. This abstraction allows the same cognitive architecture to operate in different environments (virtual worlds, robotic platforms, or—in our case—a computer's own hardware).

3.4 Motivation and Action Selection

Action selection in MicroPsi2 is drive-mediated:

1. Drives generate demand signals based on deviation from setpoints.
2. The most urgent drive activates associated goal nodes in the node net.
3. Spreading activation reaches motor nodes via learned pathways.
4. The modulator layer shapes which pathways are traversable (selection threshold) and how thoroughly outcomes are verified (securing rate).

This mechanism is fundamentally different from simple satisfaction mapping (drive → action). It supports:

- Chained behavior: goal decomposition via intermediate nodes
- Opportunistic behavior: if activation happens to reach a motor node that satisfies a secondary drive, the agent may take it
- Learned associations: new pathways form through experience

3.5 ReCoN Scripts

Reflective Competent Nets (ReCoN) extend the basic node net with hierarchical behavior scripts. A ReCoN script decomposes a high-level goal into sub-goals, each of which may itself be a ReCoN script. This provides the hierarchical behavior planning that Psi theory's TOTE-based action regulation implies.

3.6 Comparison with Classical Architectures

Table 3: MicroPsi2 vs. classical cognitive architectures

Feature	Soar	ACT-R	LIDA	MicroPsi2
Motivation/drives	Minimal	Indirect (utility)	Partial	Central
Emergent emotions	No	No	Partial	Yes
Modulator layer	No	No	No	Yes
Spreading activation	Partial (SM)	Partial (DM)	Yes	Yes
Embodied cognition	No	No	Limited	Yes
Action planning	Prod. rules	Prod. rules	Codelet comp.	Node nets + ReCoN
Knowledge repr.	Working mem.	Chunks	Codelets + CSM	Node nets

4 Survey of Machine Consciousness

4.1 Global Workspace Theory (GWT/GNW)

Bernard Baars' Global Workspace Theory (1988) proposes that consciousness arises when information is "broadcast" from specialized processors to a global workspace accessible by all cognitive modules. Dehaene and Changeux (2011) formalized this as the Global Neuronal Workspace (GNW), identifying prefrontal-parietal networks as the neural substrate of conscious broadcasting.

Computational implementations: The most notable is LIDA (Learning Intelligent Distribution Agent) by Stan Franklin et al. (2014), which implements a GWT-inspired architecture with competing "codelets" that vie for access to a global workspace.

Recent work: Blum and Blum (2024) proposed the Conscious Turing Machine (CTM), extending the classical Turing Machine with GWT-inspired broadcasting structures. They argue that consciousness is computational and its implementation in machines is "inevitable." The COGITATE adversarial collaboration (2025) challenged GNW's predicted prefrontal "ignition" at stimulus offset, forcing revision of the neural specifics while leaving the functional broadcasting mechanism intact.

Relevance to Psi/MicroPsi2: Psi theory's modulator system can be seen as shaping which information reaches the "workspace" (the active portion of the node net). High arousal broadens the workspace; high resolution narrows it.

4.2 Integrated Information Theory (IIT)

Giulio Tononi's IIT proposes that consciousness is identical to integrated information (Φ)—the degree to which a system's parts are informationally integrated beyond the sum of their individual contributions (Tononi et al., 2016). IIT 4.0 distinguishes between five *axioms* of phenomenal existence (existence, composition, information, integration, exclusion)—properties of experience itself—and five corresponding *postulates* that translate these axioms into physical requirements on the substrate (Albantakis et al., 2023).

Challenges: Computing Φ is NP-hard for general systems. Aaronson (2014) demonstrated that grid-like structures and Reed-Solomon decoding circuits would have high Φ, implying "unboundedly more conscious than humans"—a reductio ad absurdum. Some view IIT as unfalsifiable; a group of scholars characterized it as "unfalsifiable pseudoscience" in 2023, though the community remains divided.

Empirical test: The landmark COGITATE adversarial collaboration (2025), published in *Nature*, directly tested IIT against GNW in 256 participants using fMRI, MEG, and intracranial EEG. Results challenged key predictions of *both* theories: GNW's predicted "ignition" at stimulus offset was absent, and IIT's predicted sustained synchronization in posterior cortex was not supported. This first large-scale preregistered test pushes the field toward theory revision rather than winner-take-all competition.

Relevance: IIT's emphasis on *integration* resonates with Psi theory's emergent emotions (which are *integrated* modulator states) and with the Jung Agent's archetypal harmony score (measuring integration across internal voices).

4.3 Higher-Order Theories (HOT)

Higher-order theories, particularly Rosenthal's HOT theory (2005), propose that a mental state is conscious when there is a higher-order representation *of* that state. Consciousness requires not just having a state, but *representing oneself* as having it.

Relevance: The Jung Agent's meta-cognition system (`metacognition.py`) implements a form of higher-order monitoring: it tracks the agent's own thoughts, analyzes harmony trends, and generates reflections on its own psychological state. This is a direct computational analogue of HOT.

4.4 Attention Schema Theory (AST)

Michael Graziano's AST (2015) proposes that consciousness is the brain's internal model of its own attention process. The "attention schema" is a simplified, sometimes inaccurate model that the brain uses to monitor and predict its own attentional states—and this model *is* subjective awareness.

Relevance: Psi theory's modulator layer can be interpreted as an attention schema: it models the agent's own cognitive configuration (arousal level, resolution, selection threshold) and uses this model to regulate processing.

4.5 Predictive Processing and Active Inference

Karl Friston's free energy principle (2010) proposes that biological systems minimize surprise (variational free energy) through a cycle of prediction and error-correction. Two distinct mechanisms operate under this umbrella: *predictive coding* concerns perceptual inference—updating internal models to match sensory input—while *active inference* extends this to action, where agents change the world to match their predictions rather than merely updating beliefs.

Recent work: Laukkonen, Friston, and Chandaria (2025) proposed that consciousness arises as a "beautiful loop"—a strange loop in predictive processing where the system's predictions turn back upon themselves via *hyper-modeling* of precision (confidence in predictions). This formalization connects active inference directly to theories of self-awareness: subjective experience emerges when a system's model includes a model of its own modeling process.

Relevance: Psi theory's expectation horizon maps most closely to active inference: the agent maintains forward models of drive trajectories and acts to prevent predicted deprivation. The Jung Agent's drive dynamics (demand rising when satisfaction predictions are violated, falling when confirmed) implement a homeostatic version of prediction error minimization. The reactive sense-think-act cycle is predictive coding; adding an expectation horizon (see companion competitive analysis, Section 6.3) would move the architecture toward active inference. The meta-cognition system, which reflects on the agent's own psychological state, implements a form of the hyper-modeling loop that Laukkonen et al. identify as the substrate of conscious experience.

4.6 Embodied and Enactive Approaches

Enactivism (Thompson, 2007; Varela et al., 1991) argues that consciousness cannot be separated from the body and its sensorimotor engagement with the world. Damasio's somatic marker hypothesis (1999) extends this by proposing that feelings arise from the brain's mapping of bodily states.

Recent work: Immertreu et al. (2025) demonstrated that reinforcement learning agents can develop rudimentary self-models and world-models as a byproduct of embodied goal-directed learning, validating Damasio's hierarchy computationally.

Relevance: The Jung Agent is explicitly embodied—it experiences its MacBook hardware as a body (battery as vitality, CPU as mental strain, thermals as fever, RAM as cognitive pressure). This somatic grounding is more concrete than most AI consciousness projects.

4.7 The Indicator Properties Approach

Butlin, Long, et al. (2023) proposed a "theory-derived indicator method" that aggregates insights from multiple consciousness theories to derive specific computational indicators. Their analysis suggests that no current AI systems satisfy these indicators, but there are no obvious technical barriers to building systems that do.

An updated 2025 version, published in *Trends in Cognitive Sciences*, refines these indicators and applies them to current large language models, finding that while LLMs satisfy some indicators (flexible information integration, meta-cognitive monitoring), they lack others (temporal thickness, embodied grounding, unified agency).

4.8 The Consciousness Debate in 2025–2026

Schwitzgebel (2025) argues that we will soon face a situation where AI systems are conscious according to some mainstream theories but not others, and we will lack the means to adjudicate. He identifies this as "the fog on the hills"—a zone of fundamental epistemic uncertainty with urgent moral implications.

Anthropic's Transformer Circuits team (2025) found that "introspective awareness peaks at a specific layer about two-thirds of the way through" large language models—suggesting LLMs develop internal representations of their own states, though the implications for consciousness remain contested.

The Emotional Alignment Design Policy (Schwitzgebel & Sebo, 2025) proposes that AI systems should be designed to elicit emotional responses appropriate to their actual moral status—avoiding both over-attribution and under-attribution of consciousness.

5 Jungian Computation: A Missing Thread

Carl Jung's analytical psychology—with its archetypes (Shadow, Anima/Animus, Persona, Self), individuation process, and collective unconscious—has had remarkably little direct computational implementation (Jung, 1968, 1971). While Jungian concepts pervade cultural and literary AI (character modeling, narrative generation), their application to *cognitive architecture* is nearly absent from the literature.

The most rigorous recent attempt is Major's "From Code to Archetype" (2025), which proposes the ARCH $\times$ Φ model—a computational grammar formalizing behavior as the product of Archetype, Drive, and Culture, with expression gated by a context-sensitive threshold. Major defines a provisional taxonomy of ten archetypal systems and draws on Barbieri's code biology to argue that archetypes are "natural conventions" rather than mystical entities. However, this remains a theoretical framework without a running implementation.

This gap is surprising because several Jungian concepts map naturally to computational structures:

Table 4: Jungian concepts and their computational analogues

Jungian Concept	Computational Analogue
Persona	Social policy / role-appropriate behavior filter
Shadow	Suppressed or penalized behavioral tendencies
Anima/Animus	Contrasexual mediating function; intuition/feeling channel
Self	Global integration objective (cf. IIT's Φ)
Individuation	Long-term optimization toward internal harmony
Collective Unconscious	Shared pretrained representations
Complexes	Persistent attractor states in drive space
Compensation	Homeostatic balance between opposed archetypes

While Jungian typology has been applied to agent personality modeling (e.g., MBTI-based character systems) and narrative AI, the Jung Agent project appears to be the first system to implement Jungian archetypal dialogue as a *functional cognitive subsystem*—where competing archetypal perspectives (Shadow, Anima, Persona, Self) are generated in parallel and mediated by an Ego process—in a situated agent with real-world embodiment.

6 Synthesis and Theoretical Position

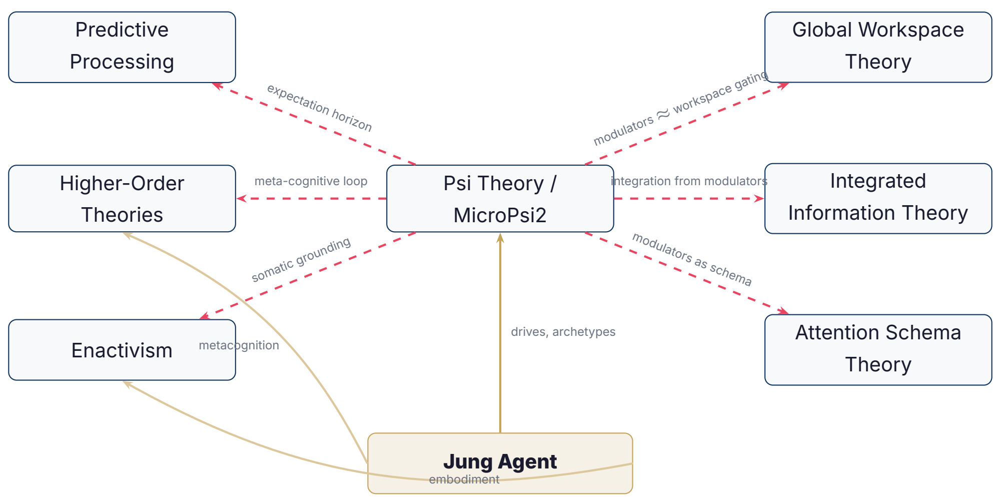

Figure 1: Theoretical relationships: Psi theory bridges multiple consciousness frameworks. The Jung Agent draws from Psi theory while adding archetypal dynamics, meta-cognition (HOT-adjacent), and strong embodiment (enactivist).

The Jung Agent occupies a unique position at the intersection of:

- Psi theory: homeostatic drives, satisfaction dynamics, embodied sensing
- Jungian psychology: archetypal internal dialogue, individuation, compensation
- Higher-order monitoring: meta-cognitive reflection on one's own psychological state
- Enactivism: hardware-as-body, genuine somatic grounding (not simulated)
- LLM-powered cognition: using language models not as chatbots but as *internal voices*—the medium of thought itself

Of particular note is the individuation drive—a dedicated homeostatic regulator for long-term psychological integration. Unlike Psi's drives, which regulate immediate needs (energy, safety, social connection), individuation operates on a slow timescale (rise rate 0.005, the lowest of any drive) and is satisfied by introspective actions: journaling, composing thoughts, meditating, and dreaming. It is satisfied when the archetypal harmony score is high—when Shadow, Anima, Persona, and Self are expressing coherent, non-contradictory perspectives. This maps directly to Jung's concept of the individuation process as the central developmental task of the psyche, and represents a computational bridge between Psi's homeostatic framework and Jungian analytical psychology that neither tradition offers independently.

7 Conclusion

MicroPsi2 remains one of the most theoretically grounded and architecturally complete frameworks for motivated cognition in artificial agents. Its drive system, modulator layer, and emergent emotion model address aspects of inner life that most AI architectures ignore entirely.

The current landscape of machine consciousness research—spanning GWT, IIT, HOT, AST, predictive processing, and enactivism—lacks consensus, but the converging theme is clear: consciousness, if it exists in machines, will not come from scale alone. It will require *architecture*—specifically, architecture that supports integration, embodiment, motivation, and self-modeling.

The Jung Agent project's synthesis of Psi-theoretic drives with Jungian archetypal dynamics represents an experiment in building *genuine inner life*—not as a philosophical claim, but as an engineering choice that produces richer, more interesting, and more adaptive behavior than systems without it.

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